

YUZE GE

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EDUCATION

Fudan University, Shanghai, China

Sep. 2023 – Present

M.S. in Statistics

GPA: 3.73/4.0

Fudan University, Shanghai, China

Sep. 2019 – Jun. 2023

B.S. in Data Science

GPA: 3.63/4.0

RESEARCH EXPERIENCE

Benefits of Learning Rate Warmup

May. 2025 - Present

Advised by Prof. Tong Zhang

Siebel School of Computing and Data Science, UIUC, USA

- We aimed to theoretically explain the benefits of learning rate warm-up from an optimization perspective.
- We proposed a novel smoothness assumption for the loss landscape of neural networks, which is strictly weaker than existing generalized smoothness assumptions like (L_0, L_1) -smoothness.
- We theoretically proved the upper bounds of gradient descent and stochastic gradient descent with a warm-up learning rate schedule, and constructed a matching lower bound, showing the benefits of warm-up schedule compared with constant learning rate.

Semismooth Newton Method for the Minimax Optimization

Jun. 2024 - Present

Advised by Prof. Rujun Jiang

Dept. Data Science, Fudan University, CN

- We applied the Semismooth Newton Method (SSN) to finding a root of a nonlinear equation defined by setting the natural residual of a convex-linear minimax problem to zero.
- We proposed efficient methods for the computation of the generalized Jacobian and Newton direction, by transforming the Newton system into a lower dimensional symmetric system.
- We proved the local error bound condition for the natural residual, and based on this we established the local superlinear convergence of SSN.

Stochastic Optimization for Spectral Risks

Dec. 2023 - Present

Advised by Prof. Rujun Jiang

Dept. Data Science, Fudan University, CN

- We designed a near optimal stochastic gradient-based algorithm for the spectral risk minimization problem, with low computational complexity.
- We transformed the spectral risk minimization problem into a minimax problem, and applied the primal-dual framework.
- We implemented the algorithm with low computational cost, achieving better gradient complexity and runtime in experiments.

Nonsmooth Nonconvex Optimization for the Rank-based Loss

Mar. 2023 - Nov. 2023

Advised by Prof. Rujun Jiang

Dept. Data Science, Fudan University, CN

- We developed a novel framework for the rank-based loss minimization problem, based on the alternating direction method of multipliers (ADMM).
- By introducing an auxiliary variable, the original nonconvex nonsmooth problem was transformed into a composite minimization problem, where each subproblem is a constrained smooth problem.
- We applied the Moreau envelope smoothing technique to analyze the convergence rate of ADMM.

PUBLICATIONS

* denotes equal contributions.

- [1] Yuxing Liu*, **Yuze Ge***, Rui Pan, Kang An and Tong Zhang. Theoretical Analysis on how Learning Rate Warmup Accelerates Convergence. [Preprint]
- [2] **Yuze Ge**, Rujun Jiang. SOREL: A Stochastic Algorithm for Spectral Risks Minimization. *International Conference on Learning Representations, 2025*. [OpenReview]
- [3] Rufeng Xiao*, **Yuze Ge***, Rujun Jiang and Yifan Yan. A Unified Framework for Rank-based Loss Minimization. *Advances in neural information processing systems, 2023*. [OpenReview]

TEACHING EXPERIENCE

Methods of Optimization (DATA130026)	Spring 2024, 2025
Teaching assistant, Advised by Prof. Rujun Jiang	Dept. Data Science, Fudan University, CN
Introduction to Artificial Intelligence (DATA130008)	Fall 2022
Teaching assistant, Advised by Prof. Zhongyu Wei	Dept. Data Science, Fudan University, CN

SCHOLARSHIP & AWARD

China National Scholarship	2024-2025
Graduate Study Scholarship, Fudan University	2023-2024
Undergraduate Merit Scholarship, Fudan University	2019-2020, 2020-2021, 2021-2022

RELEVANT COURSEWORK

Graduate Courses			
Optimization Theory	A	Foundation of Modern Probability I	A
Introduction to Algorithms	A	Data Mining	A-
Undergraduate Courses			
Stochastic Processes	A	Fundamentals of Statistics: Principles, Methods, and R	A
Methods of Optimization	A	Numerical Algorithms and Case Studies	A
Calculus on Manifolds	A	Advanced Regression Analysis (Honor)	A-
Artificial Intelligence	A	Introduction to Statistical (Machine) Learning	A-
Multivariate Statistics	A	Applied Real Analysis and Functional Analysis	A

TECHNICAL SKILLS

Programming: Python, Pytorch, MATLAB, R

SERVICE

Conference Reviewer

- International Conference on Artificial Intelligence and Statistics (AISTATS) 2025
- International Conference on Learning Representations (ICLR) 2025
- Conference on Neural Information Processing Systems (NeurIPS) 2024, 2025
- International Conference on Machine Learning (ICML) 2025